

Bank Customer Churn Prediction

Project report: exploratory analysis, decision tree and random forest classification

Notebook: Final_churn_prediction.ipynb | Author: Yash Jain

1. Introduction

When a customer closes their account, the bank loses future income, and winning a replacement customer costs more than keeping the current one. If a bank could tell in advance which customers are likely to leave, it could reach out to them first. This project looks at whether the basic information a bank already has about a customer (age, country, balance, number of products and so on) is enough to make that prediction.

I used a public Kaggle dataset of 10,000 bank customers, explored how each feature relates to leaving, and trained two classifiers, a decision tree and a random forest, and compared them on a held-out test set. Everything was done in one Colab notebook.

2. Data

The dataset (Churn for Bank Customers, from Kaggle) has 10,000 rows and 14 columns. The target is *Exited*, which is 1 if the customer closed their account and 0 if they stayed. I renamed it to *Churn*. The other columns are:

- RowNumber, CustomerId, Surname (identifiers, dropped)
- CreditScore
- Geography (France, Germany, Spain)
- Gender
- Age
- Tenure (years with the bank)
- Balance
- NumOfProducts (1 to 4)
- HasCrCard (0/1)
- IsActiveMember (0/1)
- EstimatedSalary

About 20.4% of the customers in the data left, so the classes are imbalanced. This turns out to matter for how the model results should be read.

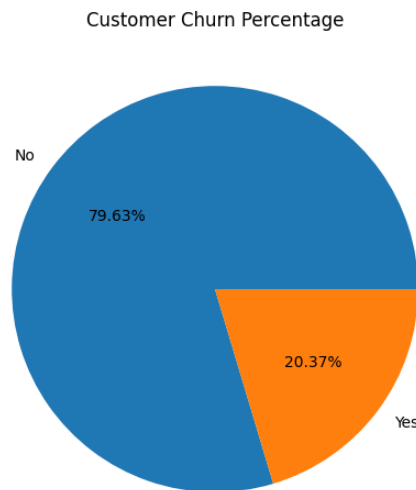


Figure 1. Share of customers who stayed (79.6%) and who left (20.4%).

3. Preprocessing

Cleaning. I dropped RowNumber, CustomerId and Surname since they are identifiers and say nothing about behaviour, and checked for null values, duplicate rows and data types before going further.

Encoding. Geography and Gender were label-encoded with scikit-learn (Geography: France 0, Germany 1, Spain 2; Gender: 0 and 1).

Scaling. CreditScore, Balance and EstimatedSalary were standardised with StandardScaler. Tree-based models do not actually need scaling, so this step made no real difference to the results.

Correlations. A correlation heatmap showed no strong relationships between the input features. The only noticeable ones with the target were Age (0.29), IsActiveMember (-0.16), Balance (0.12) and Gender (-0.11). The largest correlation between two inputs was Balance and NumOfProducts (-0.30).

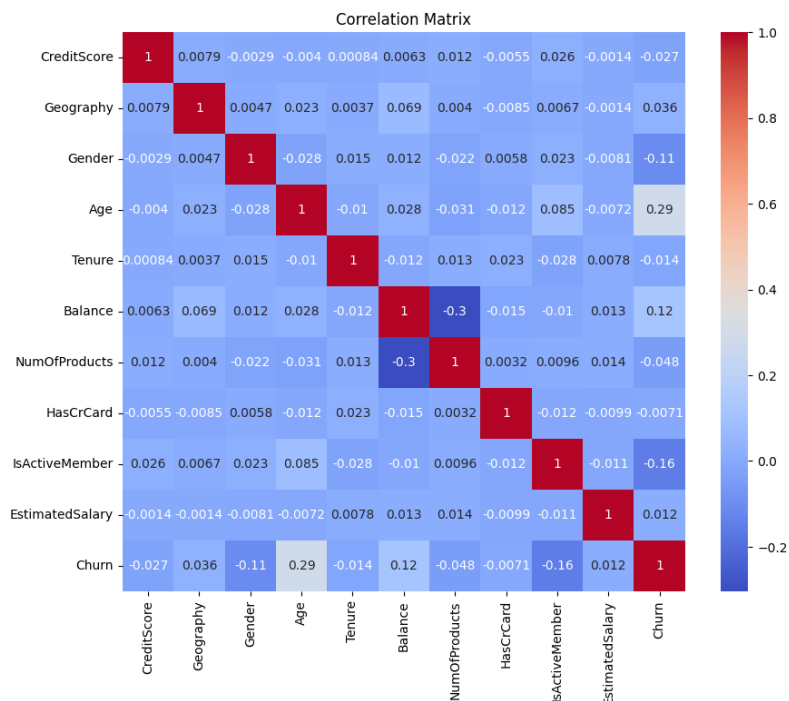


Figure 2. Correlation matrix after encoding and scaling.

4. Exploratory analysis

Gender. There are more male customers, but the female customers leave somewhat more often. The gap is not large enough to draw a firm conclusion.

Age. Most customers are 30 to 40 years old, but the number of people leaving peaks in the 40s and 50s. Age is one of the clearest signals in the dataset.

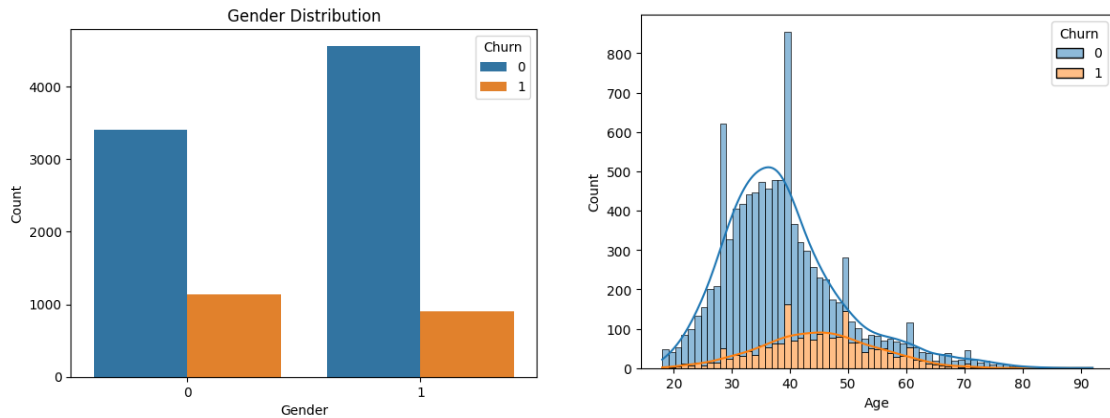


Figure 3. Churn by gender (left) and age (right).

Credit score. The box plot and violin plot look almost the same for customers who stayed and left. A few low scores appear among the leavers, but overall credit score is not a useful indicator.

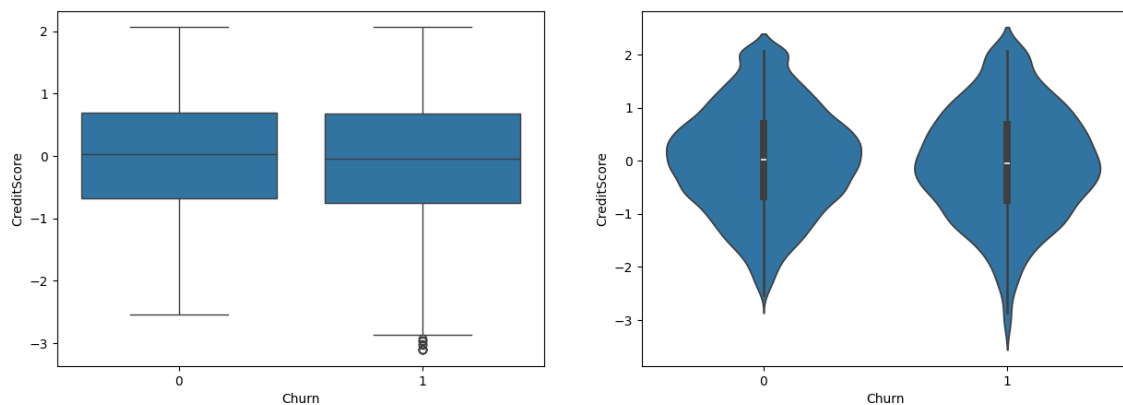


Figure 4. Credit score for customers who stayed (0) and left (1).

Geography. France has the most customers, followed by Spain and Germany. But Germany has about as many leavers as France despite having half as many customers, so German customers churn at roughly twice the rate of the other two countries.

Tenure. Most customers have between 1 and 9 years with the bank, spread fairly evenly. Very new and very long-standing customers are few, and churn is lowest for those with the longest tenure. There is no strong trend in between.

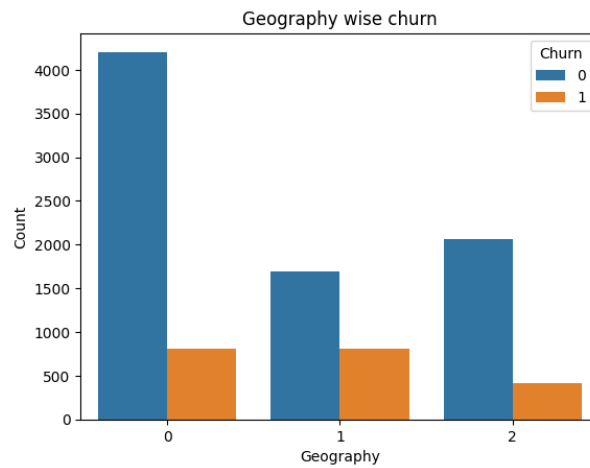


Figure 5. Churn by geography (0 = France, 1 = Germany, 2 = Spain).

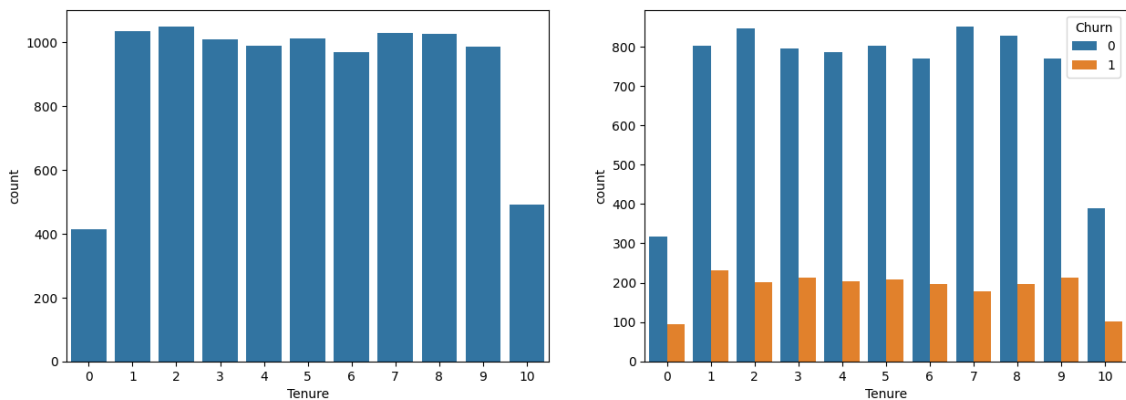


Figure 6. Tenure distribution (left) and churn by tenure (right).

Balance. A large group of customers has a zero balance. Among the rest, churn is most common for balances between 100,000 and 150,000.

Number of products. Most customers have one or two products. Those with two products have a low churn rate, while the large majority of customers with three or four products left. Customers with a single product also leave more often than those with two.

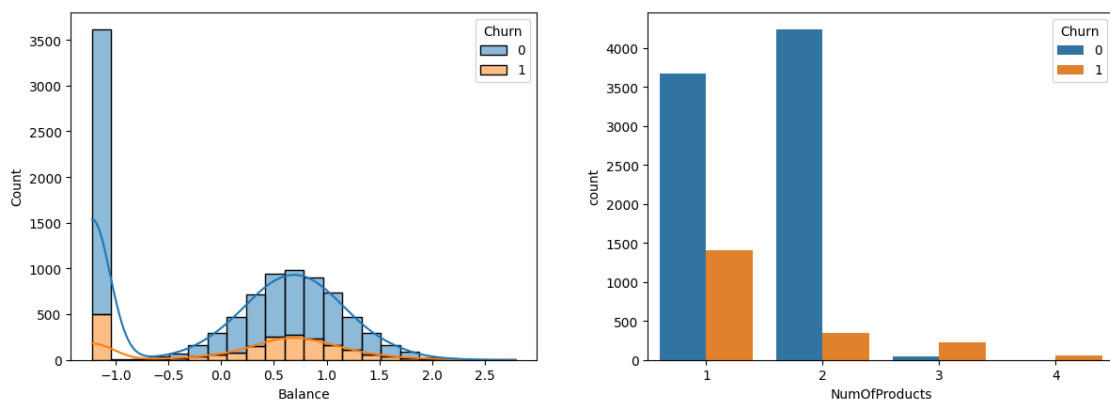


Figure 7. Churn by balance (left) and number of products (right).

Credit card and activity. About 70% of customers have a credit card, which is why more card holders appear among the leavers. Card ownership does not seem to change the churn rate (correlation with churn is about -0.007). Inactive members leave clearly more often than active ones.

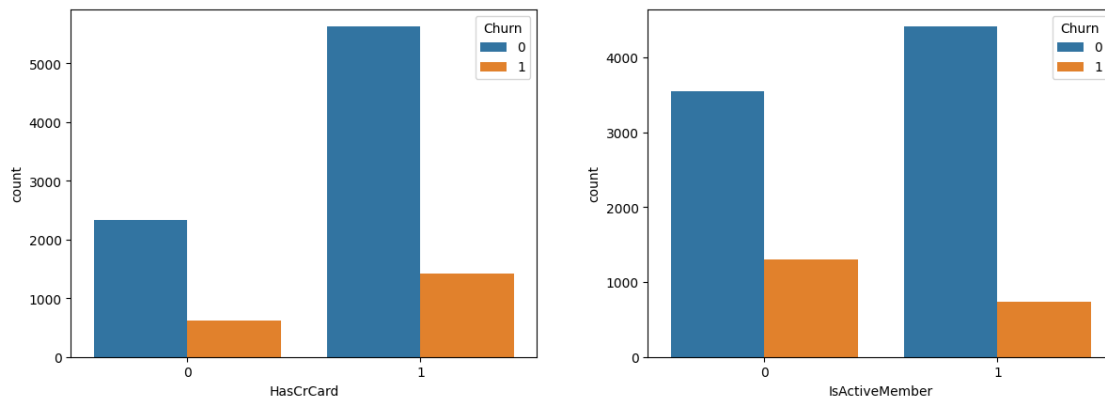


Figure 8. Churn by credit card ownership (left) and active membership (right).

Estimated salary. There is no visible pattern in salary between the two groups.

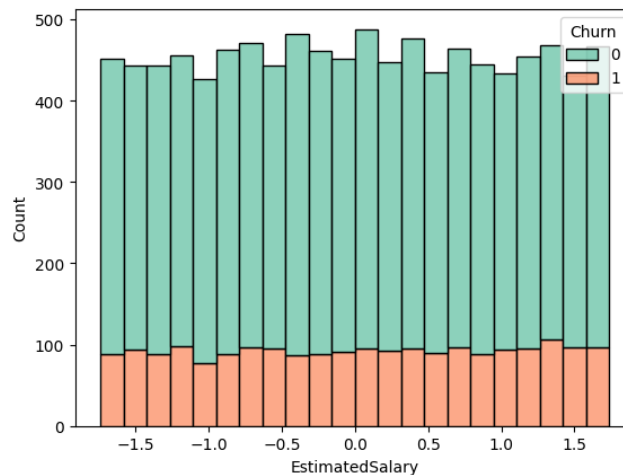


Figure 9. Estimated salary by churn.

5. Modelling

I split the data 70/30 with `random_state=42`, giving 7,000 training rows and 3,000 test rows (2,416 customers who stayed and 584 who left). Two models were trained and tuned with `GridSearchCV` using 5-fold cross-validation, scored on ROC AUC.

Decision tree

Grid: `max_depth` 2 to 20, `min_samples_leaf` 1 to 10, criterion (gini or entropy), `random_state` (0 or 42). That is 400 candidates and 2,000 fits. The best settings were gini, `max_depth` 6, `min_samples_leaf` 10.

Random forest

Grid: `max_depth` (2, 4, 6, 8, 10), `min_samples_leaf` (2, 4, 6, 8, 10), criterion, `random_state`. That is 100 candidates and 500 fits. The best settings were entropy, `max_depth` 10, `min_samples_leaf` 8, with scikit-learn's default of 100 trees.

6. Results

	Decision tree	Random forest
Train accuracy	0.858	0.877
Test accuracy	0.861	0.868
Precision (churn)	0.80	0.82
Recall (churn)	0.39	0.41
F1 (churn)	0.52	0.55
Macro F1	0.72	0.74

The random forest is better on every metric, but only by a small margin: less than one percentage point of accuracy and about three points of F1 on the churn class. Both models do very well on customers who stay (recall 0.98) and poorly on customers who leave.

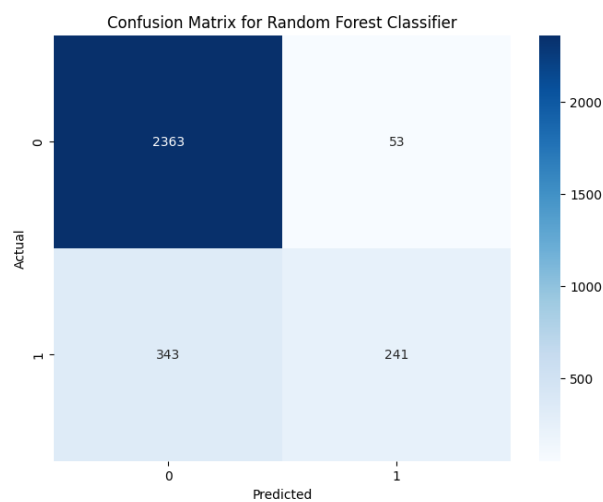


Figure 10. Confusion matrix for the random forest on the test set.

For the forest, 241 of the 584 customers who left were caught and 343 were missed, while 53 customers who stayed were wrongly flagged. In other words, when the model raises a flag it is usually right (precision 0.82), but it flags fewer than half of the people who actually leave.

It is also worth noting that a model that simply predicts "everyone stays" would score 2,416 out of 3,000, or about 80.5%, on this test set. The forest's 86.8% is better than that, but the gain is smaller than the headline number suggests.

7. Limitations

- **Low recall on churners.** The models miss around 60% of the customers who leave. For a retention use case this is the number that matters most, and I did nothing to address it (no class weights, no threshold tuning).
- **Metrics that do not fit.** The notebook also prints R^2 and mean absolute error for both classifiers (about 0.12 to 0.16 and 0.13 to 0.14). These are regression metrics and are not meaningful for a yes/no target, so I have left them out of the results above. The notebook also draws density plots of actual against predicted labels, which do not tell you much for a 0/1 variable.

- **Scaler fitted before the split.** StandardScaler was fitted on the full dataset before the train/test split, so a small amount of test-set information leaked into training. It should be fitted on the training set only.
- **Grid search details.** random_state was included as a tuned parameter, which does not make much sense since it is not a hyperparameter of the model. ROC AUC was the tuning metric, but the test-set AUC was never reported.
- **Claims that the data does not support.** My original conclusion listed Tenure and HasCrCard among the factors driving churn. The correlations for both are close to zero, so I would not rely on them. The stronger drivers are age, number of products, activity status, geography and balance.
- **One split.** All results come from a single 70/30 split, so small differences between the two models (like 0.861 against 0.868) could be down to chance.
- **No decision tree confusion matrix in the notebook output.** Only the random forest has one saved.

8. Conclusion

The exploratory analysis points to age, geography (Germany in particular), number of products, active membership and balance as the features most connected with churn. Credit score and salary show no pattern. A tuned random forest reaches 86.8% test accuracy and is slightly better than a tuned decision tree (86.1%), but neither is good at finding the customers who leave: recall is about 0.40, and the plain baseline of predicting "stays" is already around 80.5%.

So the models are a reasonable start, but I would not use them to drive retention campaigns in their current form.

9. Future work

- Handle the class imbalance with class_weight, SMOTE or a lower decision threshold, and judge the results on recall, F1 and ROC AUC.
- Fit the scaler only on the training data, or drop it for the tree models.
- Compare against logistic regression as a baseline and gradient boosting (XGBoost or LightGBM) as a stronger model.
- Use cross-validated scores rather than a single split when comparing models.
- One-hot encode Geography, and look at feature importances from the forest.
- Add interaction ideas: for example, age group combined with number of products.

10. Tools used

Python, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Google Colab.